

Hybrid Exact Category for the All-Prime Epsilon Class

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Summary

Pass 72 defines a candidate bookkeeping category $\mathcal{H}_\epsilon$ for the all-prime Loeb-Rosser class $\epsilon_{\mathbb{P}}$.

The point is to keep two pieces of information simultaneously:

1. finite conductor restricted-product support;
2. the derived pro-Ab diagonal quotient $\widehat{\mathbb{Z}}/\mathbb{Z}$.

This avoids two false simplifications:

- treating $\widehat{\mathbb{Z}}/\mathbb{Z}$ as an ordinary Hausdorff LCA quotient;
- replacing all-prime product support by finite-support characters.

The Candidate Category

An object of $\mathcal{H}_\epsilon$ has finite shadows

$$(S, k, W_{S,k}, L_{S,k}, d_S),$$

where S is a finite prime set, $k \geq 1$,

$$W_{S,k}$$

$$\prod_{p \in S} (p^{-k} \mathbb{Z}_p / p^k \mathbb{Z}_p),$$

and

$$L_{S,k}$$

$$\prod_{p \in S} (\mathbb{Z}_p / p^k \mathbb{Z}_p).$$

The boundary is

$$d_S : \mathbb{Z}^S \rightarrow \mathbb{Z}^{|S|-1}, \quad d_S(x) = (x_p - x_{p_0})_{p \neq p_0}.$$

The second layer is the lcm kernel tower

$$K_n = N_n \mathbb{Z}, \quad N_n = \text{lcm}(1, \dots, n).$$

Hybrid Exactness

A sequence is hybrid-exact if:

- all finite conductor shadows are exact;
- the pro layer is interpreted through $R^1 \varprojlim$ and supplies

$$\varprojlim^1 K_n \cong \widehat{\mathbb{Z}}/\mathbb{Z}.$$

The candidate duality is

$$\mathbb{D}_{\mathcal{H}}(d_S) = -d_S^T.$$

The pro layer is retained as derived pro-Ab data, not as a Hausdorff quotient.

Verification

The checker `code/scripts/check-pass72.py` produces `artifacts/reports/pass72-hybrid-exact-epsilon-category-check.json`.

It verifies:

- exact finite shadows for $|S| = 1, \dots, 6$;
- primitive signed-dual images;
- restriction composition for all chains among the first six primes;
- signed-dual compatibility with those restrictions;
- conductor-layer bookkeeping through $k = 1, 2, 3$;
- lcm-tower cofinality for moduli up to 24;
- non-Mittag-Leffler growth of the lcm kernel tower.

The remaining problem is external validation: prove a universal property for $\mathcal{H}_\epsilon$, or embed it faithfully into LCA sheaves, condensed/solid abelian groups, or an exact pro-category with restricted-product generators.